

Detecting Skin Trematodes: Fish Health & Behavior Surveillance Dashboard

Product Manual

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1. Product Overview

The Fish Health & Behavior Surveillance Dashboard is a computer vision tool designed to help monitor fish behavior for potential signs of *Neobenedenia* parasite infestations.

The system analyzes fish video, tracks individual fish, identifies movement patterns associated with scratching, and presents the results through a dashboard. It provides both visual and quantitative outputs to help users quickly review potential areas of concern.

The dashboard is designed so that users can run an analysis without directly interacting with the underlying computer vision code.

2. What the Dashboard Does

The dashboard combines four main stages:

1. Video Processing

Accepts raw fish footage and prepares it for analysis.

2. Fish Tracking

Identifies and tracks fish across video frames to create movement trajectories.

3. Behavior Detection

Analyzes fish trajectories and identifies movement patterns classified as "wavy," which are associated with scratching behavior.

4. Risk Assessment

Summarizes the proportion of tracked fish exhibiting wavy behavior to provide a population-level indication of potential concern.

3. Before You Begin

Before running an analysis, make sure you have:

- A fish video available for upload or stored in the designated location.
- Access to the dashboard/Google Colab environment.
- The desired analysis settings determined for your footage.

The system automatically reads information such as the video's frame rate, resolution, and duration after the video is selected.

Note: Processing can be computationally intensive and requires GPU resources, particularly for longer videos. For optimal performance, an A100 GPU or comparable high-performance GPU suited for Google Colab is recommended.

4. Dashboard Workflow

The complete workflow is:

Upload Video → (Configure Parameters*) → Run Tracking → Review Tracking Output → (Configure Parameters for Behavior Analysis*) → Run Behavior Analysis → Review Flagged Events → (Configure Parameters for risk threshold*) → Risk Assessment

***Only necessary when footage angle and position change**

Step 1: Upload a Video

Begin by uploading a **raw, untracked fish video**.

The dashboard accepts either a newly uploaded video or a video that is already available in storage.

Once the video is selected, the system automatically extracts basic video information, including:

- Frame rate
- Resolution
- Duration

The system processes the video in sequential chunks rather than attempting to process the entire video at once.

Step 2: Configure Analysis Parameters

Before processing, users can adjust the analysis parameters directly through the dashboard.

These include:

- **Video duration** — determines how much footage will be analyzed.
- **Video resolution** — determines the resolution used during processing.
- **Minimum spatial extent** — helps filter tracking results based on object size.
- **Minimum tracker confidence** — controls how confident the system must be in a tracking result.
- **Minimum visible fraction** — determines how much of a fish must remain visible for its track to be considered reliable.
- **Trail length** — controls the amount of movement history displayed for tracked fish.

These settings can be adjusted to accommodate differences in footage conditions such as camera angle, water clarity, and stocking density.

Tip: If tracking results contain substantial noise, consider using stricter tracking filters. If too many valid fish tracks are being removed, the filters may need to be relaxed.

Step 3: Run Fish Tracking

After configuring the parameters, start the tracking process.

The system processes the video in sequential segments and tracks fish throughout the footage.

The dashboard produces:

Annotated Video

The original footage is overlaid with the system's tracking results, allowing users to visually inspect how fish were detected and tracked.

Tracking Data

The system also generates a CSV containing the underlying tracking and movement information.

This data can be used for further analysis, visualization, or long-term monitoring.

5. Reviewing Fish Behavior

After tracking is complete, the behavior analysis can be run on the processed tracking data.

The system examines each fish's movement trajectory and identifies movement patterns that match the project's definition of **wavy behavior**.

Fish can be classified as:

- **Normal**
- **Wavy**

Wavy movement is used as an indicator of potential scratching behavior associated with skin irritation.

Reviewing Flagged Events

One of the main advantages of the dashboard is that it identifies the specific frames where wavy behavior was detected.

Instead of manually reviewing the entire video, users can:

1. View the detected wavy events.
2. Navigate directly to some associated frames.
3. Visually inspect the fish and surrounding footage.
4. Determine whether the flagged behavior appears consistent with scratching.

This provides a faster way to review potential events while still allowing human verification.

6. Risk Assessment

The dashboard also provides a rough population-level risk assessment.

Users can adjust the **risk threshold** and examine the percentage of tracked fish exhibiting wavy behavior.

The dashboard calculates: **Wavy Fish ÷ Total Tracked Fish**

This provides a simple measure of how widespread the detected behavior is within the analyzed group.

Important

The dashboard provides a general population-level risk assessment. The risk assessment should be treated as an **indicator for further investigation**, rather than a definitive diagnosis of parasite infestation.

Wavy movement is intended to identify potential scratching behavior, which can be associated with skin irritation and parasites.

7. Understanding the Outputs

Output	Purpose
Annotated Video	Shows fish tracking directly on the original footage
CSV Data	Contains frame-by-frame tracking and movement information
Trajectory Visualization	Shows the movement paths of tracked fish
Behavior Classification	Identifies fish/frames showing normal or wavy movement
Flagged Frames	Provides specific frames where wavy behavior was detected
Risk Assessment	Shows the proportion of tracked fish exhibiting wavy behavior

8. Interpreting Results

When reviewing results, users should consider both the quantitative output and the original footage.

A higher percentage of fish showing wavy behavior may indicate a greater level of concern, while a lower percentage may indicate less widespread abnormal behavior.

However, the mode's output should be **visually reviewed and interpreted in the context of the specific footage**.

Factors such as water clarity, camera angle, fish density, and tracking quality can affect the results.

The dashboard therefore allows analysis parameters to be adjusted for different footage conditions.

9. Troubleshooting

Fish are not being tracked accurately

Check:

- Video quality
- Water clarity
- Fish density
- Camera angle
- Tracking confidence settings
- Minimum visible fraction

Try adjusting the tracking parameters and rerunning the analysis.

Too many tracking results are being filtered out

The tracking filters may be too strict. Consider lowering the relevant filtering thresholds and rerunning the analysis.

Too many unreliable tracks are appearing

The tracking filters may be too lenient. Consider increasing the minimum confidence, visibility, or spatial requirements.

Processing is taking a long time

The pipeline uses several computationally intensive computer vision models. Longer videos and higher-resolution footage require more processing.

Consider:

- Analyzing a shorter clip
- Using a lower resolution
- Running the system with appropriate GPU resources

10. Final Notes

The dashboard is intended to make fish behavior monitoring more **consistent, objective, and efficient** than manual observation alone.

The system provides detection and quantitative information while preserving the ability for users to visually review flagged events. Its outputs can therefore serve as a starting point for identifying potential fish health concerns and determining when further investigation may be warranted.